MatchPulse — Kata

One-minute pitch

For a club organizer, MatchPulse sells named tickets and runs a points prediction market for a match day. The hard part is concurrency: capacity, balances and club data must stay correct when many fans act at once. The system is one Next.js app and one PostgreSQL database. The database enforces the rules through RLS, row locks and transactional functions. I claim that ticket sales and prediction trades stay correct under concurrent requests — no oversell, no double charge, no double payout, no cross-club leak — with the limitation that commerce is simulated and the evidence is local/CI. The detail is in the architecture description and the requirement register.

MatchPulse is a club-first football ticketing and fan-engagement platform. Clubs subscribe to a plan, create events and matches, sell named tickets, check attendees in, and read aggregate prediction insights. Fans use one account to buy tickets and predict with points across clubs. A platform admin manages clubs, plans and audit. Commerce is simulated.

Users: hundreds of fans at kickoff; 1–10 club staff (`OWNER`, `ADMIN`, `STAFF`, `ANALYST`); one platform admin/operator.

Requirements

- **R1. Club tenancy.** `STARTER`, `PRO` and `ELITE` plans gate features and limits in the database.
- **R2. Events and capacity.** Staff create events and matches; sales and registrations never exceed capacity or allocation.
- **R3. Nominal tickets.** A buyer gets 1–4 named attendees with identity, terms, a code/QR, transfer rules and idempotent check-in.
- **R4. Cross-club account.** One fan account buys and predicts across clubs; a club sees only its own data.
- **R5. Points market.** Fans enter and exit HOME/DRAW/AWAY while open; no commission, a re-entry cooldown, and score-bound one-shot resolution.
- **R6. Rewards.** Tickets earn points; fans redeem club rewards; points are not cash.
- **R7. Privacy.** Clubs see aggregate insights only; cohorts below 5 are suppressed; attendee identity is excluded.
- **R8. Administration.** Clubs, plans and audit without impersonation; CLI parity; no real payments or KYC.

Context: kickoff peaks; correctness under concurrency and retries; seat maps, wallet passes, turnstile hardware, FIGC integration and real money are out of scope.

Detail: requirements · domain model · invariant registry